Zhikai Chen

Contact

Founder @ QuantSpirit | Frontier Predictive Intelligence
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Research Interests

- Foundation models for relational data and predictive tasks (relational deep learning, in-context learning)
- LLM agents, with focus on agent security, agent data infrastructure, and post-training
- LLM infrastructure, with focus on inference and post-training systems

Selected Projects

OpenRFM: An Open Relational Foundation Model [Link]
Predictive intelligence over relational data (@QuantSpirit)

1. Building the best public relational foundation model pre-trained purely on synthetic data.
2. Pairing a relational agent, which provides the proper harness, with a predictive relational foundation model that amortizes the predictive budget while keeping strong performance.
3. Achieving performance close to KumoRFMv2, a 30% improvement over other open models.

RedCurriculum: A Curriculum-Trained Red-Teamer for Frontier LLM Agents [Link]
Post-training for red-teaming agents

1. Distributed post-training at medium scale (32+ GPUs) for a red-teaming agent, built on the SkyRL stack.
2. Designing the RedCurriculum post-training stages used to train red-teaming agents.
3. Building a red-teaming LLM that can tackle frontier LLMs such as Claude Sonnet 4.5, improving the vulnerability detection rate from 0.8% to more than 40%. It is the only model with a public training recipe that can effectively red-team frontier LLMs. Paper available soon.

Experience

Amazon AWS, Applied Scientist Intern, Santa Clara 05/2026 – 08/2026

1. Building RedCurriculum (see Selected Projects).
2. Good performance and received a return offer (currently in the team matching stage).

Amazon AWS, Applied Scientist Intern, Santa Clara 05/2025 – 12/2025

1. Developing a router module that selects the best machine learning model for a relational database, improving prediction performance by over 20% with 10x fewer hyperparameter search trials; published at ICLR 2026 (Relatron) [Link].
2. Developing a benchmark and new baseline for predictive intelligence on relational data (RFMBench) [Link], with a prototype relational foundation model recipe that outperforms existing methods.
3. Good performance and received a return offer.

Google, Student Researcher (part-time), Remote 03/2025 – 05/2025

1. Working on machine learning over relational databases (graph foundation models for relational data).

Amazon AWS, Applied Scientist Intern, Santa Clara 06/2024 – 09/2024

1. Conducting pioneering research on self-evolving agents for MLE agents, improving the data input for graph machine learning models.
2. Proposing the agent framework AutoG [Link], published at ICLR 2025; it was further turned into an open-source product at AWS and demonstrated company-wide.
3. Good performance and received a return offer.

Baidu, Research Intern, Search Strategy Team, Beijing 05/2023 – 09/2023

1. Exploring how large language models can help learning on graphs; published in SIGKDD Explorations (596 citations), with the Graph-LLM open-source repo (300+ stars) [Link].

Education

Michigan State University, East Lansing, USA 01/2023 – 2027/05 (expected)

- Ph.D., Computer Science and Engineering. Advisor: Foundation Professor Jiliang Tang

Shanghai Jiao Tong University, Shanghai 09/2020 – 01/2023
- M.Eng., Cybersecurity. Advisor: Professor Li Pan

Shanghai Jiao Tong University, Shanghai 09/2016 – 08/2020
- B.Eng., ECE, Joint Institute. Advisor: Professor Xudong Wang, IEEE Fellow

Selected Papers

1. "REDCURRICULUM: A Curriculum-Trained Red-Teamer for Frontier LLM Agents" [Link]
   Zhikai Chen et al. Under submission.
2. "OpenRFM: Dissecting Relational In-Context Learning" [Link]
   Zhikai Chen, Junyu Yin, Jialiang Gu, Siheng Xiong, Xiaoze Liu, Ruowang Zhang, Keren Zhou, Kai Guo. arXiv:2606.04320, 2026.
3. "Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline" [Link]
   Zhikai Chen, Jialiang Gu, Junyu Yin, Xianxuan Long, Shenglai Zeng, Xiaoze Liu, Kai Guo, Keren Zhou, Jiliang Tang. **IJCNLP-AACL (main)**, 2026.
4. "Position: Graph Foundation Models are Already Here" [Link]
   Haitao Mao*, **Zhikai Chen***, Wenzhuo Tang, Jianan Zhao, Yao Ma, Tong Zhao, Neil Shah, Mikhail Galkin, Jiliang Tang. **ICML**, 2024 (spotlight).
5. "Relatron: Automating Relational Machine Learning over Relational Databases" [Link]
   Zhikai Chen, Han Xie, Jian Zhang, Jiliang Tang, Xiang Song, Huzefa Rangwala. **ICLR**, 2026.
   Keywords: Relational Deep Learning, AutoML, Theory
6. "AutoG: Towards Automatic Graph Construction from Tabular Data" [Link]
   Zhikai Chen, Han Xie, Jian Zhang, Xiang Song, Jiliang Tang, Huzefa Rangwala, George Karypis. **ICLR**, 2025.
   Keywords: DSAgent, Relational Database
7. "RFMBench: Towards Principled Benchmarking of Relational Foundation Models" [Link]
   Zhikai Chen*, Han Xie, Jian Zhang, Haiyang Yu, Huzefa Rangwala, Xiang Song. Under review.
   Keywords: Relational Foundation Models
8. "Label-free Node Classification on Graphs with Large Language Models (LLMs)" [Link]
   Zhikai Chen, Haitao Mao, Hongzhi Wen, Haoyu Han, Wei Jin, Haiyang Zhang, Hui Liu, Jiliang Tang. **ICLR**, 2024.
   Keywords: LLM for Crowdsourcing, Data Annotation
9. "Exploring the Potential of Large Language Models (LLMs) in Learning on Graphs" [Link]
   Zhikai Chen, Haitao Mao, Hang Li, Wei Jin, Hongzhi Wen, Xiaochi Wei, Shuaiqiang Wang, Dawei Yin, Wenqi Fan, Hui Liu, Jiliang Tang. **SIGKDD Explorations**, 2023. (596 citations)
   Keywords: Large Language Models, Graph Machine Learning

Open-Source

1. **Amadeus** [Link] — Personal AI research assistant for reading, summarizing, organizing, and acting on academic papers. 176 GitHub stars (Sep. 2026).
2. **ML PhD Interview Notes** [Link] — Interview preparation notes for ML PhD and research positions. 59 GitHub stars (Sep. 2026).
3. **AutoG** [Link] — LLM-based data agent for automatic graph schema generation from data lakes. Internal adoption and demonstration at AWS re:Invent.
4. **Graph-LLM** [Link] — Exploring how LLMs can help learning on graphs: LLM-as-enhancer, LLM-as-predictor, LLM-as-annotator. 300+ stars.
5. **LLM-GNN** [Link] — LLMs as annotators for node classification, superior performance without human supervision. 80+ stars.

Grants and Research

Tinker Research Grant	$5,000
NSF ACCESS allocation	150,000 resource units

PUBLICATION **Conference and Journal Papers**

17. "Relatron: Automating Relational Machine Learning over Relational Databases"
Zhikai Chen, Han Xie, Jian Zhang, Jiliang Tang, Xiang Song, Huzefa Rangwala
ICLR, 2026

16. "Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline"
Zhikai Chen, Jialiang Gu, Junyu Yin, Xianxuan Long, Shenglai Zeng, Xiaoze Liu, Kai Guo, Keren Zhou, Jiliang Tang
IJCNLP-AACL (main), 2026

15. "A Pre-training Framework for Relational Data with Information-theoretic Principles"
Quang Truong, **Zhikai Chen**, Mingxuan Ju, Tong Zhao, Neil Shah, Jiliang Tang
NeurIPS, 2025

14. "Unveiling Mode Connectivity in Graph Neural Networks"
Bingheng Li, **Zhikai Chen**, Haoyu Han, Shenglai Zeng, Jingzhe Liu, Jiliang Tang
KDD, 2025

13. "Graph Machine Learning in the Era of Large Language Models (LLMs)"
Wenqi Fan, Shijie Wang, Jiani Huang, **Zhikai Chen**, Yu Song, Wenzhuo Tang, Haitao Mao, Hui Liu, Xiaorui Liu, Dawei Yin, Qing Li
ACM TIST, 2025

12. "Enhancing ID and Text Fusion via Alternative Training in Session-based Recommendation"
Juanhui Li, Haoyu Han, **Zhikai Chen**, Harry Shomer, Wei Jin, Amin Javari, Jiliang Tang
AACL (poster), 2025

11. "AutoG: Towards automatic graph construction from tabular data"
Zhikai Chen, Han Xie, Jian Zhang, Xiang Song, Jiliang Tang, Huzefa Rangwala, George Karypis
ICLR (poster), 2025

10. "Improving Causal Reasoning in Large Language Models: A Survey"
Longxuan Yu, Delin Chen, Siheng Xiong, Qingyang Wu, Dawei Li, **Zhikai Chen**, Xiaoze Liu, Liangming Pan
NAACL (main), 2025

9. "A Pure Transformer Pretraining Framework on Text-attributed Graphs"
Yu Song, Haitao Mao, Jiachen Xiao, Jingzhe Liu, **Zhikai Chen**, Wei Jin, Carl Yang, Jiliang Tang, Hui Liu
LOG (poster), 2024.

8. "Neural Scaling Laws on Graphs"
Jingzhe Liu, Haitao Mao, **Zhikai Chen**, Tong Zhao, Neil Shah, Jiliang Tang
LOG (poster), 2024.

7. "Text-space Graph Foundation Model: Comprehensive Benchmarks and New Insights"
Zhikai Chen, Haitao Mao, Jingzhe Liu, Yu Song, Bingheng Li, Wei Jin, Bahare Fatemi, Anton Tsitsulin, Bryan Perozzi, Hui Liu, Jiliang Tang
NeurIPS (poster), 2024.

6. "Position: Graph Foundation Models are Already Here"
Haitao Mao*, **Zhikai Chen***, Wenzhuo Tang, Jianan Zhao, Yao Ma, Tong Zhao, Neil Shah, Mikhail Galkin, Jiliang Tang (* Equal Contributions)
ICML (spotlight), 2024

5. "Label-free Node Classification on Graphs with Large Language Models (LLMS)"
Zhikai Chen, Haitao Mao, Hongzhi Wen, Haoyu Han, Wei Jin, Haiyang Zhang, Hui Liu, Jiliang Tang
ICLR (poster), 2024

4. "Exploring the potential of large language models (llms) in learning on graphs"
Zhikai Chen, Haitao Mao, Hang Li, Wei Jin, Hongzhi Wen, Xiaochi Wei, Shuaiqiang Wang, Dawei Yin, Wenqi Fan, Hui Liu, Jiliang Tang
SIGKDD Explorations, 2023.

3. "The 1st International Workshop on Graph Foundation Models (GFM)"
   Hosted workshop
   Haitao Mao, Jianan Zhao, Xiaoxin He, **Zhikai Chen**, Qian Huang, Zhaocheng Zhu, Jian Tang, Micheal Bronstein, Xavier Bresson, Bryan Hooi, Haiyang Zhang, Xianfeng Tang, Luo Chen, Jiliang Tang
   Companion Proceedings of the ACM on Web Conference, 2024.
2. "Demystifying Structural Disparity in Graph Neural Networks: Can One Size Fit All?"
   Haitao Mao, **Zhikai Chen**, Wei Jin, Haoyu Han, Yao Ma, Tong Zhao, Neil Shah, Jiliang Tang
   NeurIPS (poster), 2023.
1. "Fake News Detection with Contrastive Heterogeneous Graph Representation Learning"
   Zhikai Chen, Peng Wu, Li Pan
   IJCNN (poster), 2022.

Preprints and Submissions

5. "REDCURRICULUM: A Curriculum-Trained Red-Teamer for Frontier LLM Agents"
   Zhikai Chen et al. Under submission.
4. "OpenRFM: Dissecting Relational In-Context Learning"
   Zhikai Chen, Junyu Yin, Jialiang Gu, Siheng Xiong, Xiaoze Liu, Ruowang Zhang, Keren Zhou, Kai Guo
   arXiv:2606.04320, 2026
3. "RFMBench: Towards Principled Benchmarking of Relational Foundation Models"
   Zhikai Chen*, Han Xie, Jian Zhang, Haiyang Yu, Huzefa Rangwala, Xiang Song
2. "One Model for One Graph: A New Perspective for Pretraining with Cross-domain Graphs"
   Jingzhe Liu, Haitao Mao, **Zhikai Chen**, Bingheng Li, Wenqi Fan, Mingxuan Ju, Tong Zhao, Neil Shah, Jiliang Tang
1. "Learning on Graphs with Large Language Models (LLMs): A Deep Dive into Model Robustness"
   Kai Guo, Zewen Liu, **Zhikai Chen**, Hongzhi Wen, Wei Jin, Jiliang Tang, Yi Chang

Proposal Writing

- Large-scale Graph Foundation Model
  PI: Dr. Jiliang Tang, Dr. Wei Jin
  Role: Drafted part of the proposal
  Result: Computation credits funded by NVIDIA.
- Graph Foundation Models
  PI: Dr. Jiliang Tang
  Role: Designed and drafted part of the proposal
  Result: Funded by Amazon, Meta.

Presentations

Invited Talks

- "Status of graph foundation models" 05/2025
  Presentation at National University of Singapore
- "From LLM4Graph to principled graph foundation models" 02/2024
  Presentation at LOGS Seminar
- "Applying LLMs to solve graph-related problems" 10/2023
  Presentation at AITIME
- "Exploring the potential of LLMs on graphs" 08/2023
  Presentation at Shandong University

Services

Conference Reviewer

- ICML 2025, 2026
- NeurIPS 2024–2026
- ICLR 2025–2027
- SIGKDD 2023–2026

Journal Reviewer

- TKDE, TKDD, JMLR, TNNLS